# Pavel Tyapkin

## DSP / SDR & Communications Research Engineer

Ph.D. equivalent (Candidate of Technical Sciences), Radio Engineering | Currently based in Latin America | Open to relocation / remote
tjapkinpavel@gmail.com | linkedin.com/in/pavel-tyapkin-7530b4184/ | scholar.google.com/citations?user=iL0aztcAAAAJ

## Professional Summary

Communications and digital signal processing engineer with approximately **10 years of engineering experience** in satellite communications, navigation receivers, SDR, wireless systems, and embedded signal-processing platforms. Combines **MATLAB-based algorithm modeling and experimental validation** with practical implementation and debugging on **FPGA/SoC hardware**.

## Core Technical Skills

| Area | Technologies and experience |
|---|---|
| **DSP algorithms / modeling** | MATLAB, Simulink, GNU Octave; digital filtering, receiver/transmitter processing chains, multichannel reception, signal analysis and simulation |
| **Digital communications** | OFDM, BPSK/QPSK, QAM; wireless PHY processing, IEEE 802.11-based systems, satellite and navigation communication systems |
| **Interference mitigation / research** | Blind source separation, independent component analysis (ICA), FastICA, JADE, SOBI/AMUSE; interference modeling and algorithm evaluation |
| **SDR / RF experimentation** | GNU Radio, USRP B210/N210, SDR-based generation and measurement setups, RF instrumentation; experience with ADRV9009/AD9361 platforms |
| **FPGA / real-time implementation** | Verilog, Vivado, Xilinx Zynq-7000 / Artix-7; DSP-to-RTL integration, FPGA timing/resource optimization, ADC/DAC interfaces, real-time signal processing |
| **Embedded / data interfaces** | C/C++, Python, Linux, Buildroot; ARM-based platforms, AXI, DMA, PCIe/XDMA, Ethernet, SPI, I²C, UART |
| **Verification / workflow** | ModelSim, Xilinx ILA/VIO, simulation and hardware validation, Git, code review, technical documentation; AI-assisted development and research workflows subject to confidentiality requirements |

## Research Experience

### Junior Researcher — Moscow Aviation Institute (MAI)

*2023 – December 2025 · Russian Science Foundation-funded project*

- Investigated **independent component analysis and blind source separation** for suppressing interference in deep-space communication scenarios.
- Developed and evaluated signal-processing models relevant to communication links affected by interference associated with electric propulsion systems.
- Designed and built **experimental hardware testbeds and prototypes** used for investigating and validating signal processing approaches.

**Project:** *Research on improving the efficiency of deep-space communication systems using accompanying relay spacecraft in missions with electric propulsion.*
Russian Science Foundation — project information

## Professional Experience

### Lead Radio Engineer — AO Energiya Razvitiya (ER)

*October 2025 – October 2026*

- Developed and adapted FPGA-based **IEEE 802.11 wireless communication logic** using established implementation architectures.
- Integrated and evaluated PHY-oriented digital signal processing components in an FPGA/SoC environment.
- Worked on RTL integration, timing/resource optimization, radio peripheral control, and processor-accessible configuration interfaces.

*Project implementation details and quantitative results are intentionally omitted for confidentiality.*

### FPGA / Embedded Systems Developer — NII KP, Branch of ORKK JSC

*November 2020 – April 2025*

- Developed and verified **digital filters, receiver/transmitter processing chains, and communication algorithms** for satellite communication and COSPAS–SARSAT-related systems, using MATLAB, C/C++, and FPGA implementations.

- Worked on **multichannel acquisition and coherent reception**, including systems with up to eight ADC channels and signal bandwidths of approximately 50 MHz.
- Implemented and validated DSP and control logic on Xilinx FPGA/SoC platforms, integrating ADC/DAC devices, peripheral controls, and high-speed data transfer (DMA and PCIe/XDMA).

### Engineer — Topcon Positioning Systems

*September 2018 – September 2020*

- Developed DSP algorithms and FPGA/DSP logic for **navigation receiver** systems.
- Designed automated **SDR-based test and measurement systems** to characterize radio hardware and support ETSI-related testing.
- Developed signal-generation and measurement software for communication equipment validation.

### Engineering Technician — Russian Federal Space Agency

*March 2016 – June 2018*

- Participated in the development and verification of FPGA-based satellite communication systems.
- Built simulation models to study digital signal-processing stages and communication system architectures.
- Developed hardware/software testbeds for communication equipment evaluation.

## Education

**Candidate of Technical Sciences (Ph.D. equivalent), Radio Engineering — Moscow Aviation Institute, 2025**
Dissertation: *Improving the Interference Immunity of Radio Communication Systems Against Impulsive Interference Using Blind Source Separation Methods.*

**Postgraduate studies — Telecommunication Systems, Networks and Devices — Moscow Aviation Institute, completed 2024**

**M.Sc., Infocommunication Technologies and Communication Systems — Moscow Aviation Institute, 2020**

**B.Sc., Infocommunication Technologies and Communication Systems — Moscow Aviation Institute, 2018**

## Selected Publications

- **Tyapkin, P. S., Vazhenin, N. A., Plokhikh, A. P.** “Analysis of Effectiveness of Methods of Protection against Interference Emission from Stationary Plasma Thrusters in Deep-Space Communication Systems.” *Russian Aeronautics*, 2025, 68(2), pp. 501–507.
- **Tyapkin, P. S., Vazhenin, N. A., Plokhikh, A. P.** “Improvement of Interference Immunity of Deep Space Communication Systems Exposed to Interference from Stationary Plasma Thrusters.” *Russian Aeronautics*, 2024, 67(2), pp. 427–439.
- **Tyapkin, P. S., Vazhenin, N. A., Plokhikh, A. P.** “Application of Blind Signal Separation Methods in the Problems on Improving the Interference Immunity of Space Communication Systems with Quadrature Amplitude Modulation.” *Russian Aeronautics*, 2023, 66(3), pp. 615–624.

## Teaching Experience

### Teaching Instructor — Moscow Aviation Institute (MAI)

*2018 – June 2025*

- Led tutorials and laboratory sessions in **General Communication Theory**, occasionally substituting for the lecturer.

## Languages

- **Russian:** Native
- **English:** Advanced (C1, self-reported)